Chapter 30 — Background Synchronization with WorkManager

> **Part IV — Offline-First & Synchronization** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 29 (offline-first — why we sync), Ch 2 (retry/backoff), Ch 15 (idempotency), Ch 5 (Hilt — `@HiltWorker`), Ch 47 (Doze/background limits, previewed). Powers Ch 17/18 (up/downloads), Ch 31 (outbox drain), Ch 32 (delta sync), Ch 49 (notifications).
>
> **Reference surface in StreamCart:** reliably running catalog sync, outbox drain, media downloads, and analytics flush in the background — surviving app kills, reboots, and Doze.

30.1 Problem Definition and Real-World Context

Mobile apps must do work that **outlives the UI** and must run **reliably** even when the app is backgrounded or killed: syncing data (Ch 29), uploading/downloading (Ch 17/18), flushing analytics (Ch 14), refreshing tokens on a schedule. Android aggressively restricts background execution to save battery (Doze, App Standby, background-service limits, Ch 47), so naive approaches break:

- **Doing it in the UI scope.** A `viewModelScope` coroutine dies the moment the user navigates away or the process is killed — the sync/upload never completes.
- **Plain services / `AlarmManager` + wakelocks.** Modern Android kills background services and defers alarms in Doze; hand-rolled wakelock schedulers drain battery and fight the OS.
- **No guarantee across reboot/kill.** Work scheduled in memory is lost on process death or device restart.
- **Ignoring constraints.** Syncing on a metered connection or low battery wastes the user's data/power; naive code doesn't wait for Wi-Fi/charging.
- **Duplicate/overlapping work.** Triggering sync from multiple places spawns concurrent, conflicting jobs.

WorkManager is the Jetpack solution for **deferrable, guaranteed** background work: it persists work in a database, runs it respecting **constraints** (network, charging, battery), **retries with backoff**, survives **process death and reboot**, and integrates with **Doze**/background limits (Ch 47) using the right under-the- hood mechanism per API level. The architecture is about **what work goes to WorkManager vs elsewhere**, **constraints/uniqueness/backoff**, **expedited and long-running (foreground) work**, and **observing results** — getting background work to actually complete without draining the battery.

Three practical use cases

1. **Data sync (StreamCart).** A periodic + event-triggered `SyncWorker` drains the outbox (Ch 31) and pulls deltas (Ch 32) on connectivity, surviving kills/reboots.
2. **Media download (Watch).** A long-running foreground worker downloads an episode with a progress notification, constrained to unmetered/charging (Ch 18).

3. **Analytics flush / token prefetch (all).** Deferrable periodic work batched to save battery (Ch 14), expedited work for time-sensitive tasks.

Why this is architectural

Background execution defines **what work runs reliably off the UI**, how it **respects battery/network constraints and Doze**, how it **survives process death/ reboot**, and how results flow back — properties that determine whether sync, uploads, and downloads actually complete on real devices.

30.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Run deferrable work that survives app kill and device reboot.
F2	Respect constraints (network type, charging, battery not low, idle).
F3	Retry failures with exponential backoff (Ch 2).
F4	Deduplicate/uniquely name work to avoid overlap.
F5	Support periodic and one-time work; event-triggered sync.
F6	Support expedited (time-sensitive) and long-running (foreground) work.
F7	Observe progress/result from the UI; chain dependent work.

Non-functional requirements

Attribute	Policy
Reliability	Guaranteed execution (persisted); resumes after kill/reboot.
Battery/Network	Constraints + batching; respect Doze/limits (Ch 47); not polling.
Correctness	Idempotent work (Ch 15); no duplicate effects on retry.
Testability	Workers testable in isolation; DI via Hilt.
UX	Progress/result observable; foreground notification for long work.

Scope

- **In scope:** WorkManager fundamentals (Worker/CoroutineWorker, constraints, backoff, unique work, periodic/one-time, expedited, foreground/long-running, chaining, observing, `@HiltWorker`), when to use WorkManager vs alternatives, Doze interaction, testing.
- **Out of scope:** the sync *logic* (Ch 29/32), outbox (Ch 31), up/download mechanics (Ch 17/18), Doze/background-limits depth (Ch 47), foreground-service depth (Ch 46), notifications (Ch 49).

Assumptions & constraints

- WorkManager 2.9+, Hilt (`@HiltWorker` + `HiltWorkerFactory`). Idempotent work (Ch 15). `minSdk 24` . WorkManager is for **deferrable, guaranteed** work — not immediate foreground tasks nor exact-time alarms.

30.3 System Design Fundamentals

- **WorkManager = deferrable + guaranteed.** It persists work requests in a local DB and executes them via the best available mechanism per API level (JobScheduler, etc.), so work **survives process death and reboot** and runs when constraints are met. It is the recommended API for background work that must *eventually* run but not *immediately*.
- **When NOT to use it.** Immediate work tied to a visible task → a coroutine/ foreground service; exact-time triggers (alarms, calendar) → `AlarmManager` (Ch 48); in-process immediate reactions → just a coroutine. WorkManager trades immediacy for guarantees.
- **Constraints.** Declare requirements (network type, charging, battery-not-low, storage-not-low, device-idle); WorkManager defers until met — the key battery/ data lever.
- **Backoff & retry.** A worker returns `Result.retry()` to be rescheduled with **exponential or linear backoff** (Ch 2); `success` / `failure` are terminal.
- **Unique work.** `enqueueUniqueWork` / `enqueueUniquePeriodicWork` with a name + policy (`KEEP` / `REPLACE` / `APPEND`) prevents duplicate/overlapping jobs — essential for sync.
- **One-time vs periodic.** One-time for event-driven (sync-on-write, upload); periodic for recurring (min interval 15 min) — note periodic work can't be expedited and has a minimum interval.
- **Expedited work.** For time-sensitive, user-initiated work that should run ASAP (quota-limited); may require a foreground notification on some versions.
- **Long-running / foreground work.** `setForeground` / `getForegroundInfo` runs the worker as a foreground service with a notification (Ch 46) — for long tasks (large downloads, Ch 18) that need to keep running and show progress.
- **Chaining.** `beginWith(...).then(...)` builds dependency graphs (e.g., compress → upload); combine results with `WorkContinuation` .
- **Observing & DI.** `getWorkInfo*Flow` exposes state/progress to the UI; Hilt's `@HiltWorker` + `HiltWorkerFactory` injects dependencies (Ch 5) into workers.
- **Idempotency (Ch 15).** Because WorkManager may retry/re-run, the work must be idempotent (e.g., outbox mutations with keys) so re-execution is safe.

Core insight: **put deferrable, must-eventually-run work in WorkManager with the right constraints, unique naming, and backoff; make it idempotent; use foreground/ expedited for the exceptions; and observe results via `WorkInfo` .** Let the OS decide *when*, within your constraints, to maximize reliability and battery life.

30.4 High-Level Architecture



Components & ownership

- **WorkManager** persists and schedules work, enforcing constraints + backoff, surviving kill/reboot.
- **Workers** (`CoroutineWorker`, `@HiltWorker`) do the actual (idempotent) work.
- **Triggers** (write events, periodic, connectivity Ch 68) enqueue unique work.
- **Foreground** (Ch 46) for long work; **WorkInfo** flows for observation.

Direction: trigger → enqueue (unique) → WorkManager (constrained, persisted) → Worker runs → Result (retry/terminal); UI observes `WorkInfo`.

30.5 Detailed Component Design



Component	Responsibility	Notes
`CoroutineWorker` (`@HiltWorker`)	idempotent background work	`doWork(): Result`
`Constraints`	when work may run	network/charging/battery/idle
`WorkRequest`	work config	one-time/periodic, backoff, input/tags
`enqueueUniqueWork`	dedup/naming	`KEEP` / `REPLACE` / `APPEND`
`setForeground`	long work	notification (Ch 46)
`WorkInfo` flow	observe	state/progress/output

`Result` semantics: `Result.success(data)` (done, optional output), `Result.retry()` (reschedule with backoff), `Result.failure(data)` (terminal failure). Choose retry only for transient failures (Ch 2).

30.6 End-to-End Data Flow

Connectivity-triggered sync that survives a kill



Long-running download (foreground)



30.7 Complete Kotlin Implementation

Package: `com.streamcart.sync`.

30.7.1 CoroutineWorker with Hilt injection (idempotent)

```
@HiltWorker
class SyncWorker @AssistedInject constructor(
    @Assisted appContext: Context,
    @Assisted params: WorkerParameters,
    private val syncEngine: SyncEngine,              // Ch 29/32
) : CoroutineWorker(appContext, params) {

    override suspend fun doWork(): Result = try {
        when (syncEngine.sync()) {                   // idempotent (Ch 15/31)
            is SyncResult.Success -> Result.success()
            is SyncResult.Retryable -> Result.retry()        // transient -> backoff
            is SyncResult.Permanent -> Result.failure()
        }
    } catch (e: Exception) {
        if (runAttemptCount < MAX_ATTEMPTS) Result.retry() else Result.failure()
    }
    private companion object { const val MAX_ATTEMPTS = 5 }
}
```

30.7.2 Hilt WorkerFactory wiring

```
// Application implements Configuration.Provider so WorkManager uses HiltWorkerFactory.
@HiltAndroidApp
class StreamCartApp : Application(), Configuration.Provider {
    @Inject lateinit var workerFactory: HiltWorkerFactory
    override val workManagerConfiguration: Configuration
        get() = Configuration.Builder().setWorkerFactory(workerFactory).build()
}
// (and remove the default WorkManagerInitializer in the manifest)
```

30.7.3 Scheduler: unique one-time (event) + periodic, with constraints + backoff

```
class SyncScheduler @Inject constructor(@ApplicationContext private val context: Context) {
    private val wm get() = WorkManager.getInstance(context)

    private val constraints = Constraints.Builder()
        .setRequiredNetworkType(NetworkType.CONNECTED)      // wait for network (Ch 68)
        .setRequiresBatteryNotLow(true)
        .build()

    // Event-triggered (e.g., after a local write, Ch 29): unique, keep if already queued.
    fun requestSync() {
        val request = OneTimeWorkRequestBuilder<SyncWorker>()
            .setConstraints(constraints)
            .setBackoffCriteria(BackoffPolicy.EXPONENTIAL, 15, TimeUnit.SECONDS)   // Ch 2
            .addTag("sync")
            .build()
        wm.enqueueUniqueWork("sync", ExistingWorkPolicy.KEEP, request)             // dedup
    }

    // Recurring background sync (min interval 15 min); keep existing schedule.
    fun schedulePeriodic() {
        val request = PeriodicWorkRequestBuilder<SyncWorker>(6, TimeUnit.HOURS)
            .setConstraints(constraints)
            .build()
        wm.enqueueUniquePeriodicWork("sync-periodic", ExistingPeriodicWorkPolicy.KEEP, request)
    }
}
```

30.7.4 Expedited work (time-sensitive)

```
val expedited = OneTimeWorkRequestBuilder<SyncWorker>()
    .setExpedited(OutOfQuotaPolicy.RUN_AS_NON_EXPEDITED_WORK_REQUEST)   // fallback if quota
exceeded
    .build()
wm.enqueue(expedited)
```

30.7.5 Long-running foreground worker (Ch 18/46)

```
@HiltWorker
class DownloadWorker @AssistedInject constructor(/* ... */) : CoroutineWorker(ctx, params) {
    override suspend fun getForegroundInfo() = ForegroundInfo(NOTIF_ID, buildNotification(0))
    override suspend fun doWork(): Result {
        setForeground(getForegroundInfo())                // run as foreground service (Ch 46)
        // stream file; periodically: setForeground(updatedNotification); setProgress(...)
        return Result.success()
    }
}
```

30.7.6 Chaining + observing

```
// Chain: compress -> upload.
WorkManager.getInstance(context)
    .beginUniqueWork("export", ExistingWorkPolicy.REPLACE,
OneTimeWorkRequest.from(CompressWorker::class.java))
    .then(OneTimeWorkRequest.from(UploadWorker::class.java))
    .enqueue()

// Observe (UI):
val syncState: Flow<WorkInfo.State?> = WorkManager.getInstance(context)
    .getWorkInfosForUniqueWorkFlow("sync")
    .map { it.firstOrNull()?.state }
```

30.7.7 Module structure

```
:sync   SyncWorker, DownloadWorker, SyncScheduler
:app    StreamCartApp (Configuration.Provider + HiltWorkerFactory)
# SyncEngine -> Ch 29/32; Outbox -> Ch 31
```

30.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Multiple sync triggers	overlapping jobs	`enqueueUniqueWork(KEEP)` — one job
Retry re-runs work	duplicate effects	idempotent work (Ch 15); outbox keys (Ch 31)
Periodic + event work overlap	double sync	distinct unique names; idempotent
Work runs in a new process	no app state	self-contained worker; inject deps (Hilt)
`runAttemptCount` growth	infinite retry	cap attempts → `failure`
Progress updates spam	IPC overhead	throttle `setProgress`/foreground updates
Constraint flaps (network)	thrashing	WorkManager handles re-eval; backoff

- **Unique work + idempotency are the two correctness pillars:** uniqueness prevents overlap; idempotency makes retries/re-runs safe (WorkManager guarantees *at-least-once*, not *exactly-once*, execution).
- **Workers must be self-contained** (may run in a fresh process) — inject dependencies, read inputs from `inputData`, don't rely on in-memory app state.

30.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
App killed / process death	WorkManager persists	resume work on constraints	guaranteed run
Device reboot	persisted + `BOOT_COMPLETED`	re-scheduled automatically	survives reboot
Transient error (network/5xx)	exception/retryable	`Result.retry()` + backoff (Ch 2)	eventual success
Permanent error (4xx/validation)	non-retryable	`Result.failure()`; surface (Ch 29)	stop; notify
Doze / background limits	OS defers	runs in maintenance windows (Ch 47)	delayed but runs
Constraints unmet (metered/low battery)	constraint	defer until met	battery/data saved
Expedited quota exceeded	quota	fallback to non-expedited (policy)	still runs, later
Long work killed	foreground service	keeps running with notification (Ch 46)	completes
Duplicate enqueue	unique policy	`KEEP` / `REPLACE`	one job


30.9 Failure Scenarios and Recovery

> **Guaranteed execution is the headline:** once enqueued, work runs *eventually* even across kills/reboots/Doze — your job is to make it **idempotent** and **constrained**, and to return `retry` only for transient failures.

30.10 Scaling the Implementation

- **Level 1.** One `CoroutineWorker` for sync, enqueued uniquely with constraints + backoff.
- **Level 2.** Periodic + event-triggered sync; Hilt injection; observe `WorkInfo`.
- **Level 3 — production.** Expedited for time-sensitive; foreground for long work (Ch 18/46); chaining; connectivity-triggered (Ch 68); idempotent outbox drain (Ch 31).
- **Level 4 — large.** Multiple named workers per domain; remote-config sync cadence (Ch 41); adaptive constraints by network/battery; coalesced sync to avoid storms; progress UX.
- **Level 5 — enterprise.** Priority/quotas across many workers (Ch 14), sync observability/SLAs, Doze-aware scheduling policies, cross-module work governance, WorkManager + push-triggered sync (FCM, Ch 49).

Avoid overengineering: immediate in-UI work doesn't need WorkManager (use a coroutine). Use WorkManager specifically for **deferrable, must-eventually-run** work; don't force everything through it.

30.11 Alternative Approaches and Trade-offs

Need	Use	Not	Why
Deferrable, guaranteed (sync/upload)	**WorkManager**	coroutine/service	persisted, constrained, survives kill/reboot
Immediate, tied to visible UI	coroutine (`viewModelScope`)	WorkManager	no persistence needed; immediate
Immediate long user-visible task	Foreground service (Ch 46) / WorkManager foreground	background service	won't be killed; shows progress
Exact-time trigger (alarm/reminder)	`AlarmManager` (Ch 48)	WorkManager	WorkManager isn't exact-time
Push-triggered sync	FCM (Ch 49) → enqueue WorkManager	polling	server-initiated; battery-friendly
Periodic (≥15 min)	WorkManager periodic	`AlarmManager` loop	battery-efficient, constraint-aware

WorkManager vs coroutine/service: a coroutine is right for immediate work while the UI/process lives; WorkManager is right when work must survive the UI/process and run under constraints later. A foreground service is for immediate long-running user-visible work (WorkManager's foreground mode wraps this). `AlarmManager` is only for exact-time triggers. Choosing wrong → either dropped work (coroutine for sync) or battery waste/overkill (WorkManager for immediate UI work).

Periodic vs one-time + reschedule: periodic is simplest for recurring work (≥15 min, constraint-aware) but can't be expedited and drifts; one-time + manual reschedule gives finer control at more complexity. Use periodic for routine sync; event-triggered one-time for responsiveness.

30.12 Security Considerations

- **Workers run with app privileges** on possibly-sensitive data (sync, uploads) — encrypt data at rest (Ch 24); don't log PII/secrets (Ch 28) in worker output/ progress.
- **`inputData` is persisted in the WorkManager DB** — don't put secrets/tokens in `inputData`; pass ids and read secrets from Keystore-backed storage (Ch 20) inside the worker.
- **Idempotent + authorized:** synced mutations carry idempotency keys (Ch 15) and the backend authorizes each (Ch 29) — a replayed worker can't double-apply or escalate.
- **Foreground notifications** (Ch 46) for long work must not leak sensitive content in the notification text.
- **Clear scheduled work + sensitive data on logout** (Ch 22): cancel user-scoped workers; don't sync after logout.
- Client vs backend: WorkManager reliably runs client work; the backend still authorizes and validates everything the worker sends.

30.13 Performance and Resource Optimization

- **Constraints are the battery lever:** sync on unmetered/charging/battery-not-low; defer otherwise — the single biggest efficiency win.
- **Batch & coalesce:** one unique sync job (not per-change); batch outbox pushes (Ch 31) and analytics (Ch 14); avoid sync storms on reconnect.
- **Respect Doze/windows** (Ch 47): deferrable work runs in maintenance windows — don't fight it with wakelocks/alarms.
- **Minimize wakeups:** periodic cadence as low as acceptable; prefer push-triggered sync (FCM) over frequent polling.
- **Don't block:** `CoroutineWorker.doWork` is suspend; do I/O on IO dispatcher; keep foreground updates throttled.
- **Profiling:** Background Task Inspector (Android Studio) for WorkManager jobs, battery historian, sync frequency/duration, retry rate.

30.14 Testing Strategy

WorkManager provides test utilities; test workers in isolation and scheduling behavior.

```
@RunWith(AndroidJUnit4::class)
class SyncWorkerTest {
    @Before fun setUp() {
        WorkManagerTestInitHelper.initializeTestWorkManager(
            context, Configuration.Builder().setWorkerFactory(testHiltFactory).build()
        )
    }

    @Test fun `worker returns success when sync succeeds`() = runTest {
        val worker = TestListenableWorkerBuilder<SyncWorker>(context)
            .setWorkerFactory(factory(FakeSyncEngine(result = Success)))
            .build()
        assertEquals(ListenableWorker.Result.success(), worker.doWork())
    }

    @Test fun `transient failure returns retry`() = runTest {
        val worker = TestListenableWorkerBuilder<SyncWorker>(context)
            .setWorkerFactory(factory(FakeSyncEngine(result = Retryable))).build()
        assertEquals(ListenableWorker.Result.retry(), worker.doWork())
    }

    @Test fun `constraints met triggers run (test driver)`() {
        val request = OneTimeWorkRequestBuilder<SyncWorker>()

.setConstraints(Constraints.Builder().setRequiredNetworkType(NetworkType.CONNECTED).build()).bui
ld()
        val wm = WorkManager.getInstance(context); wm.enqueue(request).result.get()
        val driver = WorkManagerTestInitHelper.getTestDriver(context)!!
        driver.setAllConstraintsMet(request.id)                // simulate constraints
        assertEquals(WorkInfo.State.SUCCEEDED, wm.getWorkInfoById(request.id).get().state)
    }

    @Test fun `unique work keeps existing (no duplicate)`() { /* enqueueUniqueWork KEEP twice ->
1 */ }
}
```

What to test: `doWork` success/retry/failure (idempotent), constraint gating (test driver), unique-work dedup, backoff, periodic scheduling, foreground info. Use `TestListenableWorkerBuilder` for unit-ish worker tests, `WorkManagerTestInitHelper` for scheduling.

30.15 Production Observability

- **Metrics:** work success/retry/failure rate per worker, scheduled-vs-run latency (how long deferred), run frequency, backoff/retry counts, foreground-work completion, battery impact.

- **Logs:** `work{name, state, attempt, result, duration}` (via worker + `WorkInfo`; no PII, Ch 28), correlation IDs (Ch 1/84).
- **Debugging:** work never runs → constraints never met / cancelled / factory misconfigured (no `Configuration.Provider`); work runs repeatedly → not idempotent / wrong unique policy; battery drain → too-frequent/unconstrained work; long work killed → missing foreground.
- **Tools:** Android Studio **Background Task Inspector** shows WorkManager jobs, states, constraints, attempts.
- **Alerting:** rising work-failure rate or scheduled-to-run latency → background execution issue.

30.16 System Design Interview Preparation

> 40+ questions.

Fundamentals

1. *What is WorkManager for?* Deferrable, guaranteed background work that survives process death/reboot and respects constraints.
2. *When NOT to use it?* Immediate UI work (coroutine), exact-time triggers (AlarmManager), immediate long user-visible tasks (foreground service).
3. *Guarantee level?* At-least-once execution; survives kill/reboot; runs when constraints met.
4. *Why not plain services/alarms + wakelocks?* Doze/background limits kill/defer them; battery drain; no reboot survival.
5. *Worker types?* `Worker` (sync, threaded), `CoroutineWorker` (suspend), `ListenableWorker` (callback).

Constraints/scheduling 6. *Constraints?* Network type, charging, battery-not-low, storage-not-low, idle — defer until met. 7. *Backoff?* `Result.retry()` + exponential/linear backoff (Ch 2). 8. *Unique work?* Named work + `KEEP` / `REPLACE` / `APPEND` to avoid duplicates. 9. *One-time vs periodic?* Event-driven vs recurring (min 15 min, can't be expedited). 10. *Expedited work?* Time-sensitive, runs ASAP, quota-limited; fallback policy. 11. *Long-running work?* Foreground via `setForeground` + notification (Ch 46). 12. *Chaining?* `beginWith().then()`; `WorkContinuation`.

Correctness/reliability 13. *Why idempotent?* At-least-once execution → retries/re-runs must not duplicate (Ch 15). 14. *Survives reboot?* Yes — persisted + rescheduled. 15. *Workers run in a new process?* Can — must be self-contained; inject deps; read `inputData`. 16. *`Result` meanings?* success/retry/failure (retry = transient only). 17. *`runAttemptCount`?* Cap it → failure (no infinite retry).

DI/observing 18. *Inject deps into a worker?* `@HiltWorker` + `@AssistedInject` + `HiltWorkerFactory` + `Configuration.Provider` (Ch 5). 19. *Observe progress/result?* `getWorkInfo*Flow`; `setProgress`. 20. *Pass data?* `inputData` / `outputData` (small, persisted — no secrets).

Doze/battery (Ch 47) 21. *Doze interaction?* Deferrable work runs in maintenance windows; WorkManager handles it. 22. *Battery efficiency?* Constraints + batching + reasonable cadence; not polling. 23. *Push-triggered sync?* FCM (Ch 49) → enqueue WorkManager.

Use-case routing 24. *Sync?* WorkManager (deferrable, constrained, idempotent). 25. *Large upload/download?* WorkManager foreground (Ch 17/18). 26. *Exact reminder?* `AlarmManager` (Ch 48), not WorkManager. 27.

Immediate button action? Coroutine.

Testing 28. *How test a worker?* `TestListenableWorkerBuilder.doWork()`; assert Result. 29. *How test scheduling/constraints?* `WorkManagerTestInitHelper` + `TestDriver`. 30. *How test uniqueness?* Enqueue twice KEEP → one job.

Production debugging 31. *Work never runs?* Constraints unmet / cancelled / missing `Configuration.Provider`. 32. *Work runs repeatedly?* Not idempotent / wrong unique policy. 33. *Battery drain?* Too frequent/unconstrained. 34. *Long work killed?* Missing foreground. 35. *How inspect?* Background Task Inspector.

Design/scale 36. *Coalesce sync storms on reconnect?* Unique work + backoff + batching. 37. *Remote-tune cadence?* Remote config (Ch 41). 38. *Prioritize among workers?* Expedited + scheduling policy (Ch 14). 39. *Periodic drift/can't-expedite?* Known limitations; use one-time for precision. 40. *Hilt worker not injecting?* Missing `Configuration.Provider`/factory wiring.

Design-change challenges

1. **Sync must survive app kill + reboot.** WorkManager persisted + constraints.
2. **Large download in background.** Foreground worker (Ch 18/46).
3. **Save battery on cellular/low battery.** Constraints (unmetered/charging).
4. **Time-sensitive sync.** Expedited work + fallback.
5. **Server-initiated sync.** FCM (Ch 49) → enqueue WorkManager.

Mock walkthrough (abridged)

Clarify (deferrable? guaranteed? constraints? long?) → WorkManager vs coroutine/ alarm/service → `CoroutineWorker` + `@HiltWorker` → constraints + backoff + unique work → idempotent (Ch 15) → expedited/foreground for exceptions → observe `WorkInfo` → Doze/battery (Ch 47) → testing with test utils.

30.17 Practical Exercises

1. **Beginner.** Create a `CoroutineWorker` sync job enqueued uniquely with a `CONNECTED` constraint + exponential backoff. *Eval:* runs on network; retries transient failures; one job.
2. **Intermediate.** Wire `@HiltWorker` + `HiltWorkerFactory` + `Configuration.Provider`; observe `WorkInfo`. *Eval:* deps injected; UI sees state.
3. **Advanced.** Add a long-running foreground download worker (progress notification) and a periodic sync; make work idempotent. *Eval:* survives navigation; no duplicate effects.
4. **Production debugging.** Fix work that never runs (missing `Configuration.Provider`) and runs twice (not idempotent / wrong policy). *Eval:* runs once, correctly.
5. **System design challenge.** Design push-triggered sync: FCM wakes the app → enqueue WorkManager under constraints; coalesce with periodic. *Eval:* server- initiated; battery-friendly; no storms.

(Solutions withheld unless requested.)

30.18 Chapter Summary & Reference Sheet

Key decisions: use **WorkManager** for **deferrable, guaranteed** background work (sync, up/downloads, drains) — persisted, **constraint-aware**, surviving **kill/reboot/Doze**; make work **idempotent** (Ch 15) and **unique** (no overlap); **retry transient** failures with **backoff** (Ch 2), fail permanent ones; use **expedited** for time-sensitive and **foreground** for long work (Ch 46); inject deps via `@HiltWorker` (Ch 5); **observe** `WorkInfo`; put **no secrets in** `inputData`; **don't** use WorkManager for immediate UI work (coroutine) or exact-time triggers (AlarmManager, Ch 48).

Concepts: deferrable+guaranteed work, constraints, backoff/retry, unique work, one-time/periodic/expedited/foreground, chaining, `Result`, at-least-once + idempotency, `WorkInfo`, Doze interaction, `@HiltWorker`.

Common mistakes: sync in `viewModelScope` (dies on nav/kill); non-idempotent work (duplicates on retry); no constraints (battery/data waste); overlapping jobs (no unique work); secrets in `inputData`; using WorkManager for immediate/exact-time work; missing `Configuration.Provider` (Hilt worker won't inject); long work without foreground.

Checklist: ☐ WorkManager for deferrable/guaranteed ☐ constraints (network/charging/battery) ☐ backoff + retry (transient only) ☐ `enqueueUniqueWork` ☐ idempotent work (Ch 15) ☐ `@HiltWorker` + `Configuration.Provider` ☐ foreground for long work ☐ expedited for time-sensitive ☐ observe `WorkInfo` ☐ no secrets in `inputData` ☐ cancel user-scoped work on logout.

Key APIs: `CoroutineWorker` / `doWork` / `Result`, `OneTimeWorkRequest` / `PeriodicWorkRequest`, `Constraints`, `setBackoffCriteria`, `enqueueUniqueWork` / `enqueueUniquePeriodicWork`, `setExpedited` / `OutOfQuotaPolicy`, `setForeground` / `ForegroundInfo`, `setProgress`, `getWorkInfo*Flow`, `@HiltWorker` / `HiltWorkerFactory` / `Configuration.Provider`, `WorkManagerTestInitHelper` / `TestListenableWorkerBuilder`.

References:

- WorkManager: https://developer.android.com/topic/libraries/architecture/workmanager
- Define work / constraints: https://developer.android.com/topic/libraries/architecture/workmanager/how-to/define-work
- Long-running / foreground: https://developer.android.com/topic/libraries/architecture/workmanager/advanced/long-running
- Hilt + WorkManager: https://developer.android.com/training/dependency-injection/hilt-jetpack#workmanager
- Testing WorkManager: https://developer.android.com/topic/libraries/architecture/workmanager/how-to/testing

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; worker + Hilt wiring + scheduler (unique/periodic/expedited/foreground/chain) + observe; concurrency (unique + idempotency, new-process self-containment); failure (kill/reboot/Doze/retry); 5 levels; WorkManager-vs-alternatives and periodic-vs-one-time trade-offs; security (no secrets in inputData); perf (constraints/batching); test-util tests; observability (Task Inspector); 40 Q&A + 5 challenges. ✅
- **Expand later:** Doze/background limits (Ch 47), foreground services (Ch 46), up/downloads (Ch 17/18), outbox (Ch 31), FCM-triggered sync (Ch 49).
- **Interview coverage:** exceeds ≥40 bar.